

Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title : Script - Controlled ACL – Restrict Record Access Based on Field Value

Team Members :

- Team Member 1 – Project Lead / ACL Configuration
- Team Member 2 – ServiceNow Table & Data Configuration
- Team Member 3 – Testing & Validation
- Team Member 4 – Documentation & Demonstration

2. Project Overview

- **Purpose:** To control access to ServiceNow Institution Details records based on user roles and the Branch field.
- **Features:**
 - Role-based record access
 - Branch-based record restriction
 - READ ACL
 - CREATE ACL
 - WRITE ACL
 - DELETE ACL
 - Admin full access

3. Architecture

- **Platform:** ServiceNow
- **Access Control:** Script-Controlled ACL
- **Table:** Institution Details (u_institution_details)
- **Security:** Roles and Branch field-based access control
- **ACL Operations:** Read, Create, Write and Delete

4. Setup Instructions

- **Prerequisites:** ServiceNow instance / ServiceNow Lab Instance
- **Setup:**
 1. Create the Institution Details table.
 2. Create the required fields.
 3. Create the required roles.
 4. Assign roles to users.

5. Create the Script-Controlled ACLs.
6. Create Institution Details records with different Branch values.
7. Test access using different users.

5. Folder Structure

Since this is a **ServiceNow configuration project**, a MERN-style Client/Server folder structure is not applicable.

Project components:

- Institution Details Table
- User Roles
- Read ACL
- Create ACL
- Write ACL
- Delete ACL
- Test Users
- Test Records

6. Running the Application

- Open the ServiceNow instance.
- Login with the required test user.
- Navigate to **Institution Details**.
- Perform the required record operation.
- Verify access according to the assigned role and Branch value.

7. API Documentation

No custom REST API is used in this project.

The project uses ServiceNow's built-in table and ACL mechanisms to control record access.

8. Authentication

ServiceNow user authentication is used.

Authorization is controlled using:

- bb1 – READ access
- bb2 – CREATE access
- bb3 – WRITE access
- bb4 – DELETE access
- admin – Full access

9. User Interface

Include screenshots showing:

- Institution Details list

Table

New record

Submit

Cancel

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label

Institution details

* Name

u_institution_details

Extends table

Application

Global

Create module

Create mobile module

Add module to menu

-- Create new --

New menu name

Institution details

Remote Table

Columns

Controls

Application Access

Table Columns

for text

Search

Dictionary Entries

- EEE records

servicenow

AllFavoritesHistoryWorkspacesUser - New Record

Search

User

Save

Configure >Export >Create Favorite >Copy URL >Copy sys_id >Reload form

To save the record, save the record and then click Set Password.

ID

EEE User

Name

EEE

Lastname

User

Title

Department

Password needs reset

Locked out

Active

Identity type

Human

Internal Integration User

Email

eeeuser@gmail.com

Language

-- None --

Calendar integration

Outlook

Time zone

System (America/Los_Angeles)

Date format

System (yyyy-MM-dd)

Business phone

Mobile phone

Photo

Click to add...

Submit

- Read record

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record ⓘ

* Operation: read ⓘ

Decision Type: Allow If

Application: Global ⓘ

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None -- Tables

* Name: Institution Details [u_institution_details] -- None --

Description:

Applies To: No. of records matching the condition: 4
 Add Filter Condition Add OR Clause

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

- Create Record

servicenow All Favorites History Admin : Access Control - New Record Search

Access Control New record Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record ⓘ

* Operation: delete ⓘ

Decision Type: Allow If

Application: Global ⓘ

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Applies To: No. of records matching the condition: 4
 Add Filter Condition Add OR Clause

- Edit record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: delete

Decision Type: Allow if

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Applies To: No. of records matching the condition: 4

Add Filter Condition Add OR Clause

- Delete record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: delete

Decision Type: Allow if

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Applies To: No. of records matching the condition: 4

Add Filter Condition Add OR Clause

10. Testing

Testing covers:

- READ access
- Unauthorized READ access
- Admin access
- CREATE operation
- WRITE operation
- DELETE operation
- Multiple user access
- Record loading and ACL response

All completed test cases passed.

12. Known Issues

No known issues were identified during the completed testing.

13. Future Enhancements

- Add more department/branch-based access conditions.
- Add more granular role-based permissions.
- Extend ACL rules to additional ServiceNow tables.
- Add more complex field-based access conditions.
- Improve reporting and monitoring of access attempts.